Are ABRA's null results real? A test for hidden team heterogeneity

2026-07-30. Prompted by VGC-Bench (arXiv 2506.10326), which measures policy performance collapsing as the team pool grows: agents trained on 3 teams score 21% in-distribution, on 10 teams 17%, on 30 teams 8%. ABRA's pool is **4,885 teams** — more than a hundred times the point at which the published work falls apart.

That raised a specific worry. This week produced four null results in a row. If a single fixed policy cannot be good across thousands of teams, a change that helps on some teams and hurts on others would read as a clean null in the aggregate — and the aggregate is the only thing this project has ever looked at.

The hypothesis, and the version of it that was wrong

The tempting framing is that the nulls are *drowned in team variance*. **That framing is wrong, and it is worth being explicit about why**, because it nearly cost four hours of compute.

`paired_h2h.js` already removes team bias by construction. Each matchup is played twice, once with each policy on each side. If a matchup is lopsided enough that the team decides it, the same *team* wins both games — which means different *policies* won them — and the pair is recorded as a SPLIT and discarded. Splits are not averaged in. Team diversity costs **sample size**, not validity.

So the honest question is not "is the estimate biased." It is not. The question is:

> *Is there one effect here, or many effects with different signs that happen to cancel?*

The test

Under the null that every team shares one true win rate, per-team decisive wins are binomial and the spread of per-team rates is entirely predicted by the counts. Real heterogeneity shows up as **overdispersion** — more spread than binomial allows.

Pearson's chi-square across teams measures exactly that. With many teams it is approximately normal with mean `df` and standard deviation `sqrt(2·df)`, giving a z-score for "more spread than chance." An overdispersion ratio of 1.00 means the spread is precisely what luck predicts.

A team is a six, keyed by sorted species. Each decisive pair is evidence about both sixes on the table. Teams with fewer than 8 decisive pairs are excluded rather than left to add noise as singletons. Implementation: `engine/team_heterogeneity.js`.

Positive control

The test has to be shown to detect heterogeneity before its silence means anything. The greedy experiment is the strongest effect this project has measured (+12 points of raw win rate), and it should not be uniform across teams — taking your best move matters more with some teams than others.

run	decisive pairs	overall	overdispersion	z
greedy vs sampling	24,153	79.7%	**1.169**	**7.4**

Detected, clearly. Per-team rates run from 100% down to 20%. The instrument works.

Result

run	decisive pairs	overall	overdispersion	z	verdict
forced-replacement lever	17,074	50.5%	**1.005**	0.2	one shared rate
switch features (switching on)	7,097	50.5%	**1.011**	0.1	one shared rate

Both nulls are real nulls. There is no subgroup of teams on which the post-KO replacement scorer or the new switch-in features are quietly winning. They do nothing, uniformly, everywhere.

The lever result is the well-powered one — 2,098 teams and 20,227 team-observations, and an overdispersion of 1.005 is about as flat as this measurement can come out. The switch-features run is weaker evidence on its own: only 46 teams cleared the 8-pair threshold, for 388 observations, so that test had little power to find heterogeneity even if it existed. Directionally it agrees, but it should not be leaned on alone.

What this rules out, and what it does not

Ruled out: that this week's nulls are a measurement artifact of team diversity. They are not. The planned follow-up — re-running the null experiments on a narrowed team pool of 10 or 30 teams — would have cost about four hours of compute to confirm something the existing data already answers. It is cancelled.

Not ruled out: VGC-Bench's actual finding, which is about *capacity*, not measurement. They trained separate policies per team-set and watched in-distribution performance collapse as the set grew. ABRA has never tried that. A policy refitted on ten teams and tested on those same ten teams might beat the general policy by a lot, which would say the linear model is capacity-limited across the metagame and argue for conditioning the policy on team features. That is a real open question and a different experiment.

Where this leaves the week

Four feature-addition experiments, four nulls, and now confirmation that the nulls are genuine rather than hidden structure. Against that, one intervention — playing the best-scoring move instead of sampling — is worth twelve points.

The pattern is consistent and it is not about knowledge. It is about what the policy is aimed at and how it is used. VGC-Bench's own result points the same way: behaviour cloning beat pure reinforcement learning, and every one of their top agents was behaviour cloning **followed by population-based self-play improvement**. ABRA has the behaviour clone (`fit_policy`) and it has the self-play farm (MEW, 40 games/second). It has never connected them — the farm only ever measures, it has never once trained the policy.

That loop is the recommendation, ahead of more features.